Point-by-Point Response to Reviewers

TUSCO Manuscript

March 9, 2026

Contents

1 Reviewer #1 **3**
- 1.1 Comment 1: Distinction between TUSCO-novel and TUSCO-decoy . . . 3
- 1.2 Comment 2: Types of isoform transcriptome reconstruction . . . 4
- 1.3 Comment 3: Length biases in long-read sequencing protocols . . . 5
- 1.4 Comment 4: Very deep datasets . . . 5
- 1.5 Comment 5: Example TUSCO report . . . 6
- 1.6 Comment 6: Universal set download on website . . . 6

2 Reviewer #2 **7**
- 2.1 Comment 1: Statistical testing for SIRVs vs TUSCO . . . 7
- 2.2 Comment 2: Reference annotations for novel transcript evaluation . . . 8
- 2.3 Comment 3: FLAIR collapse step . . . 9

3 Reviewer #3 **10**
- 3.1 Comment 1: Define benchmarking goals . . . 10
- 3.2 Comment 2: Comparison with short-read baseline approach . . . 11
- 3.3 Comment 3: False negatives in disease studies . . . 14
- 3.4 Comment 4: Statistics interpretation . . . 15
- 3.5 Comment 5: Radar plots criticism . . . 16
- 3.6 Comment 6: TUSCO transcript length enrichment . . . 17
- 3.7 Comment 7: TUSCO terminology consistency . . . 17
- 3.8 Comment 8: “Biologically plausible” definition . . . 18
- 3.9 Comment 9: Parameter selection justification . . . 19
- 3.10 Comment 10: Cosine similarity interpretation . . . 21
- 3.11 Comment 11: TP coverage variability . . . 22
- 3.12 Comment 12: Per-read coverage definition . . . 23
- 3.13 Comment 13: Code availability . . . 24

4 Reviewer #4 **25**
- 4.1 Comment 1: Section 2.4 - Correlation interpretation . . . 25
- 4.2 Comment 2: Section 2.5 - TUSCO-novel evaluation . . . 26
- 4.3 Comment 3: Section 2.6 - Replicate number recommendations . . . 27

5 Reviewer #5 **28**

Introduction

We thank the reviewers for their thoughtful and constructive feedback. We have addressed each point and revised the manuscript accordingly. Below are our point-by-point responses.

1 Reviewer #1

Reviewer's Overall Assessment

In this manuscript, Liu and colleagues present a framework to test long-read transcriptome assembly pipelines and underlying datasets, based on a set of "simple" genes that are well-expressed and with a single isoform. These sets are carefully chosen for human and mouse, and the accompanying website is very easy to navigate. The TUSCO benchmarking framework is integrated into SQANTI3, a popular QC tool, which will make its adoption easy. **The work is logical, well supported and presented very clearly.**

Documentation is excellent.

Response

We appreciate the reviewer's positive assessment and the kind words about the documentation.

1.1 Comment 1: Distinction between TUSCO-novel and TUSCO-decoy

Reviewer Comment

The distinction between TUSCO-novel (annotation from those genes is removed) vs TUSCO-decoy (a misleading annotation is added in its place) should be made explicit, with results presented for both (along with the results when the full annotation is provided). It's debatable whether it's "fair" to provide inaccurate annotation. Minimap2 uses splice junction annotation for mapping so the alignments may already be biased before entering isoform discovery.

Response

We apologize for the confusion—there is **no separate "TUSCO-decoy" approach**. What we previously referred to as a "decoy" is part of TUSCO-novel: it removes the true transcript from the annotation and replaces it with an artificial variant. To avoid ambiguity, we have removed the term "decoy" from the revised manuscript.

In TUSCO-novel, for each single-isoform gene in the TUSCO gene set, we generate an artificial transcript with modified splice junctions using canonical motifs (GT/AG) found in the genome sequence. This artificial transcript replaces the true annotation entirely. Tools must then reconstruct the correct (but now unannotated) isoform from the sequencing reads despite the presence of this misleading reference. See the Methods (lines 632–648) for technical details. We chose to replace rather than simply delete annotations because, while gene-level annotations are largely complete in human and mouse, isoform-level annotations are often incomplete. This design mirrors a common real-world setting: the gene is known, but the splice pattern may be novel (e.g., tissue-, disease-, or condition-specific isoforms where annotations are unreliable). Simply deleting the annotation would not test this capability.

Fig. 4b presents results for both conditions in a dumbbell-plot format, directly comparing each metric under the correct reference annotation versus under TUSCO-novel. A quantitative summary is provided in Supplementary Table 3.

Regarding minimap2 splice junction bias, we share this concern. To prevent annotation bias from affecting our evaluation, *we deliberately excluded all reference annotations during the minimap2*

alignment stage. Reads were aligned using only the genome sequence (FASTA), not gene models, ensuring alignments reflect genuine sequence evidence rather than annotation guidance. This design choice, while reducing initial alignment sensitivity compared to typical workflows, ensures that TUSCO-novel evaluates end-to-end, data-driven isoform discovery—the capability needed when working with incomplete or misleading annotations.

Changes in Manuscript: We have updated the abstract to clarify the TUSCO-novel approach and removed the term "decoy" to avoid confusion (lines 30-33):

> "Masking transcripts in the TUSCO gene set and replacing them with modified splice variants in the annotation creates a TUSCO-novel challenge that assesses reconstruction of the true, now-unannotated isoforms."

1.2 Comment 2: Types of isoform transcriptome reconstruction

> **Reviewer Comment**
>
> Please make explicit what types of isoform transcriptome reconstruction can be assessed with TUSCO: genome-guided de novo (no annotation provided), genome-guided discovery (annotation present, look for new transcripts), genome reference-free (no genome or transcriptome prior information)?

Response

The TUSCO benchmarking framework is agnostic to the strategy adopted by the transcript reconstruction algorithm and could, in principle, support TUSCO evaluation of all three reconstruction types. However, the TUSCO gene set is presently only defined for human and mouse, both organisms with high-quality genomes and annotations. Hence, TUSCO evaluation is currently applicable only when tested on human and/or mouse data. Future developments will seek to expand the TUSCO gene set to other species.

Changes in Manuscript: We have added a sentence in Section 2.2 ("TUSCO benchmarking framework in SQANTI3", lines 187–194) to explicitly state this capability:

> "Note that the TUSCO benchmarking framework is agnostic to the reconstruction strategy used by a transcript reconstruction algorithm. In principle, TUSCO evaluation can assess all three major classes of transcriptome reconstruction tools: genome-guided de novo (no annotation provided), genome-guided discovery (annotation provided, with the goal of identifying novel transcripts), and genome reference-free (no genome or transcriptome prior information). TUSCO evaluation can assess any such method as long as it is applied to human or mouse data, the two species for which the TUSCO gene set is currently defined."

1.3 Comment 3: Length biases in long-read sequencing protocols

> **Reviewer Comment**
>
> All the long-read sequencing protocols have strong length biases (PacBio against short or long transcripts, ONT cDNA against even moderately long transcripts, ONT RNA against longest,...); with log scales it's hard to figure out the relevant transcript lengths in relation to these biases. Consider binning into <600 bp, 600-1kb, 1kb-2kb, . . . 4kb+ or similar.

Response

We agree that log scales can obscure specific length biases relevant to long-read protocols. We have revised Fig. 2c and Supplementary Fig. 2 panels a–f to display transcript length distributions in discrete bins (<600 bp, 600–1kb, 1kb–2kb, 2kb–4kb, >4kb) as suggested. This visualization more clearly demonstrates how transcripts in the TUSCO gene set span the relevant length categories compared to GENCODE annotations and spike-in controls.

Changes in Manuscript: We have updated Fig. 2c and Supplementary Fig. 2 panels a–f and their captions to present the transcript length distributions in the suggested bins instead of the log10 scale. Additionally, we have revised the text in Section 2.1 (lines 139–142) to refer to the new binned visualization.

1.4 Comment 4: Very deep datasets

> **Reviewer Comment**
>
> Would very deep datasets (100M reads?) cause issues, because rare transcription noise would still pass the threshold as a new isoform?

Response

We agree that very deep sequencing (e.g., 100M reads) captures more rare events, but whether this "noise" is reported as novel isoforms depends heavily on the filtering strategies of the reconstruction tools:

- **IsoQuant (Relative Thresholds):** IsoQuant filters novel transcripts that do not exceed 2% of the maximum graph coverage. As sequencing depth increases, the absolute number of reads required to meet this 2% threshold rises proportionally. Since stochastic transcription noise typically remains at low absolute frequencies, it fails to cross this dynamic threshold even in deep datasets.
- **Bambu (Probabilistic Modeling):** Bambu groups reads into "Read Classes" based on shared splice patterns and employs probabilistic modeling to identify high-confidence isoforms. Rare, stochastic noise that does not form consistent, high-confidence read classes is effectively filtered out during this consolidation process.
- **Iso-Seq + SQANTI3 (ML Filtering):** The SQANTI3 Machine Learning filter uses a Random Forest classifier trained on structural and quality attributes (e.g., TSS/TTS

support, canonical junctions) to distinguish true isoforms from artifacts. This classifier learns from the features of annotated transcripts, which include expression levels, and it may call a rare transcript as true depending on how similar their features are to those of the annotated transcripts.

If a tool fails to filter out this transcriptional noise and reports it as an isoform, TUSCO evaluation will identify it as a False Positive (FP), as genes in the TUSCO gene set are strictly single-isoform. Thus, TUSCO evaluation remains a valid benchmark for assessing a tool's ability to control for noise in deep datasets.

1.5 Comment 5: Example TUSCO report

> **Reviewer Comment**
>
> Consider providing an example report of TUSCO.

Response

We now include an example HTML report generated by the TUSCO benchmarking framework. It illustrates the diagnostic plots and QC metrics the framework produces, and shows how users can assess long-read data quality and reconstruction performance. The example TUSCO report is available at: `https://github.com/ConesaLab/SQANTI3/blob/master/example/tusco_output/tusco_report.html`. This reference is included in the Data and Code availability section (lines 767–768).

1.6 Comment 6: Universal set download on website

> **Reviewer Comment**
>
> On the TUSCO website, it's easy to download the tissue-specific gene sets but I couldn't see how to download the universal set. It can be found on the SQANTI3 Github but it would be convenient to put it on the TUSCO website too.

Response

We have updated the TUSCO website (tusco.uv.es) to include a direct download link for the universal TUSCO gene set, alongside the tissue-specific sets.

2 Reviewer #2

Reviewer's Overall Assessment

The authors report on the TUSCO benchmarking framework, a newly created benchmark for evaluating the accuracy of transcript reconstruction with long RNA sequencing reads (LR). The TUSCO gene set consists of transcripts encoded in the human or mouse genome and therefore captures both technical and biological aspects that are not captured by artificial spike-in datasets (Sequins, SIRVs) or simulation models, such as RNA degradation and extraction artifacts.

The manuscript is well-written and substantiated, with in-depth evaluations using the TUSCO benchmarking framework that demonstrate its advantages. The construction of the 'novel variant' test set is ingenious.

The primary drawback of the TUSCO benchmarking framework is that it cannot be used to assess transcript reconstruction in the presence of multiple isoforms per gene, which is one of the main promises and goals of long read RNA sequencing and LR transcriptomic analyses. The authors already recognize this as a limitation and address it in their Discussion section.

Nevertheless, overall, **the TUSCO benchmarking framework is the first endogenous benchmark that is organically integrated with the experiment, and the only one to reflect the complexities in LR RNA datasets resulting from RNA degradation and RNA capture. Therefore, it represents an original and valuable complement and addition to the benchmark sets for LR RNA sequencing.**

Code availability: While I did not assess the current SQANTI3 code, I was able to locate the general (cross-tissue) datasets and access the web site with the tissue-specific transcripts based on the instructions and links provided.

Response

We appreciate the reviewer's positive assessment of the TUSCO benchmarking framework and the note on code/data availability; we are glad the datasets and website were easy to find.

2.1 Comment 1: Statistical testing for SIRVs vs TUSCO

Reviewer Comment

Section 2.4 and Fig. 3b: When comparing the performance metrics for TUSCO and SIRVs, the authors mention that no significant differences were found for the TP and FN categories, whereas the figures suggest otherwise, i.e., SIRVs overtaking TUSCO for TPs. This could be because the statistical test was formulated for 'TUSCO > SIRVs', for which the null hypothesis could not be rejected. The authors should incorporate testing for 'SIRVs > TUSCO'.

Response

Thank you for pointing this out. We revised this analysis to explicitly test both directions using a two-sided paired sign-flip permutation test.

This result is expected and informative. TUSCO evaluation shows significantly higher rates of Partial True Positives (PTP) and False Negatives (FN) compared to SIRVs (two-sided paired sign-flip permutation test, $n = 6$ paired pipelines per species; human PTP $p = 0.046$, FN $p = 0.046$;

mouse PTP $p = 0.046$, FN $p = 0.046$). The corresponding lower TP rate (human $p = 0.046$; mouse $p = 0.015$) is driven by these increases in PTP and FN rather than an explosion of FP, which are not significantly elevated (human FP $p = 0.077$; mouse FP $p = 0.877$).

This difference arises because genes in the TUSCO gene set are endogenous and thus subject to biological processes like RNA degradation (leading to PTP) and protocol-specific limitations (leading to FN). In contrast, synthetic SIRV spike-ins are added at high concentrations and are not subject to extraction-induced degradation. We have demonstrated these effects in our RNA degradation analysis (Fig. 3c), where the fraction of fully reconstructed transcripts in the TUSCO gene set ($\frac{TP}{TP+PTP}$) correlates strongly with RIN (Pearson $R = 0.810$, $p = 4.60 \times 10^{-5}$), whereas Sequins showed no correlation ($R = 0.075$, $p = 0.77$). FN rates are further shown to be dominated by platform/protocol choice in Supplementary Fig. 3. Therefore, the lower TP rate in TUSCO evaluation accurately reflects the true sample quality and protocol limitations, providing a more realistic benchmark than the artificially inflated performance seen with SIRVs.

Changes in Manuscript: We have updated Section 2.4 (lines 229–237) to explicitly report this significant difference using two-sided tests and explain that it reflects the sensitivity of TUSCO evaluation to sample quality factors that SIRVs do not capture, citing our specific analyses as evidence:

> "Across both species, spike-ins showed higher TP rates than TUSCO evaluation (two-sided paired sign-flip permutation test across six pipelines, $n = 6$ pairs; human TP $p = 0.046$, mouse TP $p = 0.015$). Closer inspection of the lower TP for TUSCO evaluation (Fig. 3b) shows that it is driven by increases in PTP and FN rather than an explosion of FP: PTP and FN are higher under TUSCO evaluation (human PTP $p = 0.046$, FN $p = 0.046$; mouse PTP $p = 0.046$, FN $p = 0.046$), whereas FP are not elevated (human FP $p = 0.077$, mouse FP $p = 0.877$). We hypothesized that PTP arises from RNA degradation causing truncated molecules, while FN reflects protocol limitations."

2.2 Comment 2: Reference annotations for novel transcript evaluation

> **Reviewer Comment**
>
> In evaluating the programs' (Bambu, StringTie2, FLAIR, Iso-Seq+SQANTI3 ML) performance for 'novel' transcripts, it is not clear what reference annotations were provided to the programs to create the radar plots. More specifically, did the annotations include all genes and variants in GENCODE, or only those related to the TUSCO gene set? Also, how were transcripts predicted by the programs at loci outside of the TUSCO gene set accounted for?

Response

We appreciate the reviewer's request for clarification regarding the reference annotations used in the 'novel' transcript evaluation. For the benchmarking in Fig. 4, the reconstruction programs (Bambu, StringTie2, FLAIR, IsoQuant) were provided with a comprehensive reference annotation that included all genes and isoforms in GENCODE, with a specific modification restricted to the loci in the TUSCO gene set. At these loci, the true biological transcript models

were removed and replaced by artificial variants with entirely different splice patterns, using canonical donor and acceptor sites at all junctions. This modified full-genome annotation (e.g., `gencode_tusco_novel_hsa.gtf`) served as the input guide for the genome-guided tools. For the Iso-Seq pipeline (official PacBio Iso-Seq pipeline + SQANTI3-ML filtering with complementary short-read sequencing data), this modified annotation was used during the SQANTI3 QC and Machine Learning filtering steps.
Transcripts predicted by the programs at loci outside of the TUSCO gene set were excluded from the performance metrics calculation. As detailed in the "Methods" section under "TUSCO and SIRV metrics calculation", our evaluation was strictly restricted to reconstructed transcripts that mapped to genes in the TUSCO gene set, as these constitute our ground truth (identified by matching gene symbols or IDs in the SQANTI3 classification). Consequently, predictions at other genomic loci did not influence the sensitivity, precision, or other metrics. as shown in Fig. 4.

2.3 Comment 3: FLAIR collapse step

> **Reviewer Comment**
>
> Section 4.3, Transcriptome reconstruction with FLAIR. Was the 'collapse' step run with or without a reference annotation?

Response

The FLAIR "collapse" step was run **with a reference annotation** provided. As described in Section 4.5 (Transcriptome Reconstruction with FLAIR, lines 658–662), alignment files were refined with the FLAIR `correct` subcommand to adjust splice junctions and transcript boundaries based on an external reference GTF annotation. The `collapse` step was also run with the same reference annotation provided via the `--gtf` flag. This approach ensures that transcript boundaries are guided by known annotations while still allowing for novel isoform discovery based on read evidence.

Changes in Manuscript: We have clarified this in the Methods section (Section 4.5, Transcriptome Reconstruction with FLAIR, lines 658–662):

> Alignment files were refined with the FLAIR `correct` subcommand to adjust splice junctions and transcript boundaries based on an external reference GTF annotation. The `collapse` step, which groups corrected reads into isoforms, was also run with the same reference annotation provided via the `--gtf` flag, with Illumina short-read support applied only in the LRGASP long+short (LS) dataset.

3 Reviewer #3

Reviewer's Overall Assessment

To improve the benchmarking of long read sequencing (LRS) methods, the authors present the TUSCO benchmarking framework (Transcriptome Universal Single-isoform Control). The TUSCO benchmarking framework includes a computational pipeline applied to varied RNA-seq data which selects endogenous, widely-expressed, single-isoform genes to serve as internal controls.

The TUSCO manuscript presents important ideas which are worthy of publication. The strategy of defining internal controls via expressed single-isoform genes is compelling, and the authors make good arguments for the utility of this approach. However, the current manuscript suffers from major gaps that should be addressed prior to publication.

Response

We appreciate the reviewer's positive assessment and constructive feedback.

3.1 Comment 1: Define benchmarking goals

Reviewer Comment

At a high-level, what is this work about? The authors state it is about "benchmarking long reads transcriptomics", though what exactly is involved in such benchmarking is never defined. While this is implicitly addressed later by the stats the authors derive, the goals of benchmarking should be clearly stated/defined.

Response

Thank you for this suggestion. The goal of the TUSCO benchmarking framework is to evaluate the accuracy and completeness of long-read transcript reconstruction by measuring:

1. **Sensitivity (detection completeness):** Whether transcripts known to be expressed are successfully detected by the reconstruction pipeline.
2. **Precision (structural accuracy):** Whether detected transcripts match the expected reference structure without spurious splice sites or artificial variants.
3. **Robustness to sample quality:** How well pipelines perform under real-world conditions including RNA degradation, sequencing depth variations, and library preparation artifacts.
4. **Novel isoform discovery capacity:** The ability to detect true transcripts even when reference annotations are incomplete or misleading.

Changes in Manuscript: We have added an explicit statement of benchmarking goals in Section 2.2 ("TUSCO benchmarking framework in SQANTI3", lines 153–158):

> "The TUSCO benchmarking framework aims to evaluate four key aspects of long-read transcript reconstruction: (1) sensitivity: whether expressed transcripts are detected;

(2) precision: whether detected transcripts match expected structures; (3) robustness: performance under real-world sample quality variations; and (4) novel discovery capacity: ability to identify true transcripts with incomplete annotations."

3.2 Comment 2: Comparison with short-read baseline approach

> **Reviewer Comment**
>
> In the abstract the authors postulate benchmarking has been done so far by two approaches: Spiked-in transcripts and synthetic data generation. The entire manuscript is focused on comparing to spiked-in transcripts to show the TUSCO benchmarking framework advantage. However, these are definitely not the only approaches. Specifically, recent work (MAJIQ-L, Han et al, Genome Research 2024) used short-reads sequenced from the same samples as a baseline to compare different long reads algorithms and long reads technology (ONT vs PacBio). While on the surface these are different evaluations (full isoforms reconstruction is not possible with short reads) in practice much of what the authors aim to assess (and what users care about) is totally achievable by analyzing short read sequences. Specifically, while transcript start/end are not available directly from short reads, Han et al pipeline allows users to assess where long reads report junctions, splice sites, and intron retention that do not exist by short reads (or annotation). Clearly, there could be advantages and disadvantages to assessing long reads with TUSCO evaluation vs. using short reads. For example, short reads require the matching experiment to be done with short reads and is not suitable for assessing start/end of transcripts. It also uses short-reads as the baseline which, by definition, cannot be considered an absolute truth but more of an approximation with its own biases. Nonetheless, using short read carry important potential benefits: It can be done for any condition easily and will not be sensitive to the definition of synthetically introduced transcripts in the TUSCO gene set or to actual splice variants occurring in genes in the TUSCO gene set in a condition/disease. It also gives a much broader "natural" set of genes with all their built in complexity without resorting to special case genes which may be biased in some unknown way (more on that later). It also gives a wider spread of TPM, gene length etc. to assess performance, important factors glossed over in the current manuscript. Consequently, it may offer much more accurate assessment of false negatives (see below).
> While the above qualitative assessment may be sensible it is also speculative and not quantitative. Given the importance of this to improving long reads transcriptomics benchmarking (the stated goal of this manuscript), the authors must investigate this thoroughly. Indeed Han et al used much of the same data from the long reads consortium and all their analysis/results are available. Thus, it should be relatively straightforward to compare TUSCO on the same data. The fact that Han et al used this orthogonal approach/method (MAJIQ based analysis) with the same data can help shed more light on TUSCO.

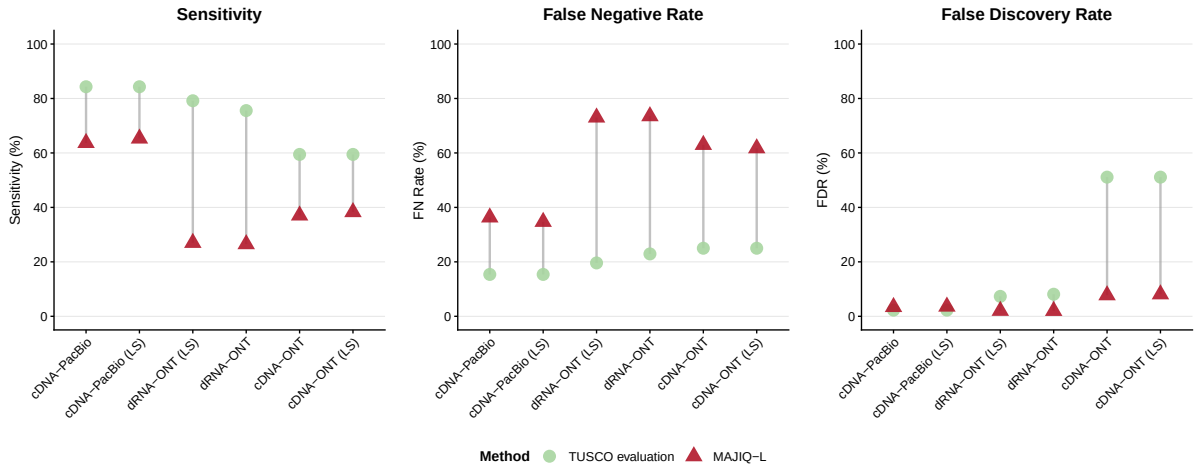
Response

We thank the reviewer for this suggestion.

We agree that short reads have historically been used to support and validate long-read transcript models. For instance, FLAIR incorporates short-read evidence to identify novel-not-in-catalog transcripts, and SQANTI3 employs short-read support as a key QC feature. Short-read–based benchmarking can indeed increase the number of transcript models available for evaluation, although this expansion is inherently limited to transcripts containing unique junctions that short reads can confidently resolve. At the same time, short-read benchmarking introduces well-recognized limitations. Because short reads lack contiguity, they cannot directly validate transcript-level structures, leaving open the question of whether long-read–derived transcript models—especially novel or recombined ones—are correctly reconstructed end-to-end. As the

reviewer notes, short reads also cannot reliably assess TSS or TTS, which remain among the most challenging features of long-read RNA-seq to benchmark. Looking forward, as long-read sequencing technologies continue to increase in throughput and decrease in cost, matched short-read datasets may become less common, reducing the practicality of relying on short reads for benchmarking in many experimental settings.

As a consequence, **TUSCO evaluation** and **MAJIQ-L** have fundamentally different scopes and target distinct dimensions of transcriptome reconstruction quality: **MAJIQ-L** focuses on *junction-level* comparison, identifying where long reads report junctions, splice sites, and intron retention that differ from short-read evidence. This is excellent for detecting junction-level errors but uses short reads as an approximation of truth rather than an absolute ground truth. In contrast, **TUSCO evaluation** focuses on *full-length transcript reconstruction*, including TSS/TTS boundaries that short reads cannot assess. TUSCO evaluation provides an absolute ground truth (single-isoform genes with no evidence of alternative splicing) rather than a relative baseline. Therefore a direct comparison is not straightforward. Nevertheless, we agree with the reviewer that a quantitative comparison against the short-read–based approach of Han et al. (MAJIQ-L, Genome Research 2024) is interesting, particularly because their study used much of the same LRGASP material. We have therefore performed this analysis. Specifically, we re-processed the datasets using the same versions of the tools evaluated in our study to ensure methodological consistency. We then compared TUSCO metrics with those derived from the MAJIQ-based short-read benchmarking approach across all tools included in our manuscript.
The results of this comparative analysis are now included in the revised version of the manuscript. We have performed a systematic comparison of the benchmarking metrics that can be computed with TUSCO evaluation versus MAJIQ-L, and present the results below (Supplementary Fig. 5).



Supplementary Fig. 5. Comparison of TUSCO evaluation and MAJIQ-L benchmarking results. Left panel: Sensitivity (TP / (TP + FN)). Middle panel: False Negative rate (FN / (TP + FN)). Right panel: False Discovery Rate ($100 \times (N - TP)/N$). Green circles represent TUSCO evaluation metrics; red triangles represent MAJIQ-L metrics. Lines connect the two methods for each pipeline. Only cDNA-PacBio consistently ranks best in both methods; dRNA-ONT and cDNA-ONT show divergent rankings that reflect different quality dimensions (RNA integrity vs. sequencing depth).

Metric Definitions

To facilitate a fair comparison, we first define how benchmarking metrics are computed with each approach in a manner comparable to the TUSCO benchmarking framework's definitions. For MAJIQ-L, junctions are classified as **known** (present in the annotation) or **novel** (not in the annotation):

Metric	Known Junctions	Novel Junctions
Ground truth	All junctions identified by MAJIQ	All novel junctions identified by MAJIQ
TP	TP-known: LR known + MAJIQ	TP-novel: LR novel + MAJIQ novel
FP	FP-known: LR known, no MAJIQ	FP-novel: LR novel, no MAJIQ
TN	*Not possible (unknown expression)*	*Not possible by MAJIQ*
FN	FN-known: No LR, Yes MAJIQ	FN-novel: No LR, Yes MAJIQ novel

Note that **TN cannot be computed** with MAJIQ-L because we do not know if a gene is expressed.

False Discovery Rate (FDR) is computed as: FDR = $100 \times (N - TP)/N$, where N is the number of observed transcripts (TUSCO evaluation) or LR junctions (MAJIQ-L). For TUSCO evaluation, N = TP + PTP + FP (observed transcripts); for MAJIQ-L, N = total_LR_junctions and TP includes both known and novel.

MAJIQ-L Benchmarking Results

We applied the MAJIQ-L framework to the same LRGASP WTC11 datasets. Supplementary Table 2 shows the junction-level metrics aggregated across all expressed genes:

Supplementary Table 2. MAJIQ-L junction-level benchmarking results.

Pipeline	TP-known	FP-known	FN-known	TP-novel	FP-novel	FN-novel
cDNA-ONT	80,021	4,397	124,611	839	2,385	12,847
cDNA-ONT (LS)	81,343	4,447	123,289	2,221	2,904	11,465
cDNA-PacBio	138,081	3,617	66,551	939	1,312	12,747
cDNA-PacBio (LS)	139,903	3,637	64,729	2,655	1,645	11,031
dRNA-ONT	57,801	971	146,831	52	208	13,634
dRNA-ONT (LS)	58,718	980	145,914	181	237	13,505

From these values, we compute Sensitivity as (TP-known + TP-novel) / (TP-known + TP-novel + FN-known + FN-novel), FN Rate as (FN-known + FN-novel) / (TP-known + TP-novel + FN-known + FN-novel), and FDR as $100 \times (N - TP)/N$ where N is the total number of LR junctions detected.

Comparison of Benchmarking Insights

We compared TUSCO evaluation with MAJIQ-L, a junction-level benchmarking approach using matched short-read data as a proxy reference (Supplementary Table 2). Among LRGASP library preparation workflows, cDNA-PacBio consistently ranked best under both frameworks, while cDNA-ONT and dRNA-ONT rankings diverged (Supplementary Fig. 5). This divergence

reflects complementary quality dimensions: TUSCO evaluation is sensitive to RNA integrity (detecting partial transcripts), whereas MAJIQ-L reflects sequencing depth across all expressed junctions. For example, dRNA-ONT achieves high TUSCO evaluation sensitivity (75.6%) despite only 760K reads, but shows a high MAJIQ-L false-negative rate (73.5%) due to limited depth. Conversely, cDNA-ONT has 16.6M reads and better MAJIQ-L junction sensitivity (37.0%) but lower TUSCO evaluation sensitivity (59.5%) due to degraded RNA. These results demonstrate that TUSCO evaluation and MAJIQ-L provide complementary assessments of long-read transcriptome reconstruction quality. Overall, pipeline ranking was similar between TUSCO evaluation and MAJIQ-L benchmarks, although MAJIQ-L results indicated lower overall sensitivity and higher FN rates, while TUSCO evaluation highlighted FDR limitations unseen by MAJIQ-L.
The two methods differ fundamentally in their approach. MAJIQ-L evaluates junction-level metrics and can assess novel junctions not present in the annotation, while TUSCO evaluation evaluates transcript-level metrics including full transcript structure and is limited to consistently expressed genes in the TUSCO gene set. A key advantage of TUSCO evaluation is its ability to evaluate TSS and TTS accuracy through the PTP (partially true positive) category, which MAJIQ-L cannot assess. Furthermore, TUSCO evaluation provides a true ground truth based on the reference annotation, whereas MAJIQ-L uses short-read data as an approximation of the truth.

Conclusion

TUSCO evaluation and MAJIQ-L capture different aspects of long-read transcriptomics performance. TUSCO evaluation provides a true annotation-based ground truth rather than relying on short-read approximations, can evaluate both PTP and FN at the transcript level, requires no additional sequencing costs (no matched short-read data needed), and uniquely captures TSS and TTS accuracy through the PTP category.
Changes in Manuscript: The revised version of the manuscript includes these analyses and conclusions in the Introduction (lines 80-86), Section 2.4 (Results, lines 266-281), Section 3 (Discussion, lines 412-423), Section 4.6 (Methods, lines 663-681), Supplementary Fig. 5, and Supplementary Table 2.

3.3 Comment 3: False negatives in disease studies

> **Reviewer Comment**
>
> The claim that TUSCO evaluation can detect false negatives even in unusual biological conditions like disease studies is suspect. While the genes in the TUSCO gene set were selected based on an analysis of many cell types and other factors, transcript expression is known to vary widely in diverse conditions, especially disease. This point should be reframed or better argued in the manuscript.

Response

We appreciate this important concern and agree that transcript expression can vary substantially in disease contexts. Some genes in the TUSCO gene set may be downregulated or absent in specific diseases, potentially inflating apparent FN rates. However, we believe that our method remains robust for detecting false alternative transcript events, as the absence of alternative

splice sites in genes in the TUSCO gene set has been extensively evaluated through our selection pipeline.

Still, there are a number of aspects that TUSCO evaluation can assess in experiments involving disease:

- **Accuracy on control samples:** Disease experiments often involve healthy control samples that still serve as a suitable testbed for method benchmarking.
- **Sample quality indicators:** The PTP/TP ratio provides a measure of RNA integrity and degradation that applies regardless of disease state.
- **Pipeline performance baseline:** Even in disease samples, TUSCO evaluation provides a baseline for comparing different reconstruction pipelines applied to the same data.

Changes in Manuscript: We have added the following text to the Discussion (lines 452–459):

> Moreover, while genes in the TUSCO gene set are selected for consistent expression across healthy tissues, their expression profiles may be altered in unusual biological conditions such as severe disease states. In these contexts, specific genes in the TUSCO gene set might be downregulated or absent, potentially inflating the apparent false negative rate. Consequently, users should interpret TUSCO metrics in disease samples with appropriate caution, using them primarily as relative indicators for comparing pipeline performance on the same dataset rather than as absolute measures of sensitivity.

3.4 Comment 4: Statistics interpretation

> **Reviewer Comment**
>
> Are the statistics used to assess performance supposed to represent "absolute truth" of some kind? Relative ranking of methods/algorithms by specific metrics? This is unclear and definitions/statements need to be made and supported accordingly. Clearly there are major effects of coverage, transcript length (see also point below) and the variance observed in the reported statistics is so high it leaves the reader wondering what does this all mean. For example, PTP average in mouse (p. 8 top) is 54% with 43% std (!).

Response

TUSCO metrics serve two complementary purposes:

1. **Relative ranking of methods/pipelines:** For a given dataset, TUSCO evaluation enables comparison of different reconstruction tools (e.g., Bambu vs. FLAIR vs. Iso-Seq) by providing consistent metrics across all tools. This ranking is the primary intended use.
2. **Sample quality assessment:** Within a single pipeline, TUSCO metrics (particularly PTP rates and TP/PTP ratios) indicate sample-specific factors like RNA integrity, which can vary substantially.

The high variance observed (e.g., PTP average ∼54% with 43% std in mouse) reflects *real biological and technical heterogeneity* across samples, which is precisely what TUSCO evaluation is designed to capture. This variance arises from:

- Differences in RNA integrity (RIN) across samples (demonstrated in Fig. 3c)
- Sequencing depth variations (demonstrated in Supplementary Fig. 3)
- Library preparation method differences (ONT vs. PacBio, dRNA vs. cDNA)
- Biological variation in expression levels

We agree the variance can be large, but it is expected—it reflects the sensitivity of TUSCO evaluation to real experimental factors. By contrast, SIRVs show artificially low variance (Supplementary Fig. 6) precisely because they bypass sample preparation variability.

Changes in Manuscript: We have added clarifying text in Section 2.3 (lines 218–221) explaining that TUSCO metrics should be interpreted as: (1) relative performance rankings within a dataset, and (2) indicators of sample-specific quality, with expected variance reflecting real technical heterogeneity.

3.5 Comment 5: Radar plots criticism

> **Reviewer Comment**
>
> Radar polygons and radar area are used throughout the manuscript for visualization and tool comparison. While these may be visually pleasing we see this as a case of form over function. Radar area is not an appropriate statistic to summarize and compare method performance (what does it actually mean?). Worse, the calculated area value depends strongly on the clockwise order of statistics; for example, for the six metrics shown in Fig. 3a, hypothetical scores ordered 1, 1, 1, 0, 0, 0 (nonzero area) have a larger radar area than 1, 0, 1, 0, 1, 0 (zero area).
>
> One wonders whether conclusions would change with a different ordering. Even in human perceptions there may be additional biases (e.g. things on the top vs on the sides). Radar polygon plots are not an intuitive way to compare tool performance because it is virtually impossible to visually compare each individual statistic across tools. The manuscript should use standard visualization methods (e.g. bar charts) allowing for comparison between statistics. It's simply hard to assess the authors' claims and converge to conclusions with these plots.

Response

Changes in Manuscript: We agree with the reviewer's critique regarding the limitations of radar plots. We have replaced all radar plots in the manuscript with dumbbell plots (Figures 3, 4, and 5) and removed radar-area-based summaries. This visualization enables direct comparison of each statistic across tools and conditions, independent of metric ordering.

3.6 Comment 6: TUSCO transcript length enrichment

> **Reviewer Comment**
>
> Section 2.1 makes the claim that transcripts in the TUSCO gene set are comparable in length to RefSeq transcripts (Fig. 1c), but this is not actually supported by Fig. 1c, which illustrates an apparent enrichment of longer transcripts in the TUSCO gene set (even more-so given the log scale in Fig. 1c). We can envision a different manuscript showing exactly the same plots as in Fig. 1c,d and making the *apposite* claim i.e. that they are significantly different, attaching a p-value to that claim...
>
> What are the implications of using unusually long transcripts as the internal controls? Moreover, this fact is a bit surprising and should be explained, as one would not necessarily expect single-isoform genes to generate relatively long transcripts. This points to a needed discussion about what biases are introduced by using such strongly-curated transcripts as the internal controls. Coming back to points raised above

Response

The reviewer correctly observes that transcripts in the TUSCO gene set tend toward the longer end of the distribution. This enrichment reflects the biological characteristics of single-isoform genes: many encode core cellular machinery, including mitochondrial translation factors, RNA-processing components, protein trafficking regulators, and metabolic enzymes, that require larger, multi-domain proteins. Because these genes are dosage-sensitive and functionally constrained, they are under strong selective pressure to maintain a single canonical transcript, which correlates with greater transcript length.
For benchmarking, this length distribution has both advantages and limitations. Longer transcripts present greater challenges for full-length reconstruction, making TUSCO evaluation a stringent test of pipeline performance. However, performance on short transcripts (<600 bp) may be underrepresented.

Changes in Manuscript: We revised Fig. 2c to display binned length distributions (<600 bp, 600–1kb, 1kb–2kb, 2kb–4kb, >4kb) for clearer interpretation, and added the following acknowledgment in Section 2.1 (Results, lines 139–142):

> Compared with reference annotations and spike-ins, the TUSCO gene set spans endogenous size/structure ranges relevant for full-length detection: transcripts are comparable to RefSeq transcripts and longer than SIRVs/ERCCs.

3.7 Comment 7: TUSCO terminology consistency

> **Reviewer Comment**
>
> Throughout the manuscript, TUSCO is confusingly referred to as a set of genes, a framework, and a collection of analyses. To this reviewer, TUSCO is certainly more than a set of genes since it includes the upstream computational pipelines and analysis pipelines, which are significant contributions. We recommend using terms more specifically to refer to the full collection of pipelines and results vs. specific parts like the cross-tissue filtered gene sets.

Response

Clarification of terminology: We thank the reviewer for this suggestion. We have standardized terminology throughout the manuscript:

- **"TUSCO gene set"**: The curated collection of single-isoform genes used as ground truth.
- **"TUSCO benchmarking framework"**: The complete system including gene selection pipeline, SQANTI3 integration, and metrics calculation.
- **"TUSCO evaluation"**: The process of applying the TUSCO benchmarking framework to assess a transcriptome reconstruction.
- **"TUSCO metrics"**: The specific performance measurements (TP, PTP, FP, FN rates, sensitivity, precision, etc.).
- **"TUSCO-novel"**: The evaluation mode using modified annotations.

Changes in Manuscript: We have revised the manuscript to consistently use these terms throughout. This includes standardizing terminology in the abstract, introduction, results sections, and methods. The key changes involve replacing ambiguous uses of "TUSCO" with the specific qualified term (e.g., "TUSCO gene set", "TUSCO benchmarking framework", "TUSCO evaluation", or "TUSCO metrics") to avoid confusion.

3.8 Comment 8: "Biologically plausible" definition

> **Reviewer Comment**
>
> The TUSCO-novel claim of "biologically plausible isoforms" (section 2.5) is interesting, but the validity of "biologically plausible" is unclear. Were the made-up splice sites chosen to e.g. preserve splice motifs / enhancers, etc? What about exon length and GC content bias?

Response

In the current TUSCO-novel simulator implementation, the artificial isoforms are designed to be *structurally plausible* in the limited sense of satisfying basic splice-junction sequence constraints. For multi-exon transcripts, each annotated intron is modified by shifting the donor and acceptor coordinates *within the original intron bounds* while searching the gene's genomic sequence for canonical dinucleotides: `GT` at the donor site and `AG` at the acceptor site on the plus strand, and the corresponding reference-strand motifs `AC` (donor) and `CT` (acceptor) on the minus strand. The search is directional (donor downstream / acceptor upstream on the plus strand) with a default window of 100 bp, falling back to scanning the full intron if needed, and it enforces a minimum shift of 5 bp when a suitable motif can be found. For single-exon transcripts, the simulator introduces one synthetic intron by selecting two positions separated by at least 50 bp and then adjusting each boundary to a nearby canonical motif within 20 bp when possible.
That said, several aspects of *biological* plausibility are not modeled or guaranteed. Splice enhancer/silencer sequences are not preserved, branch point / polypyrimidine tract features are not

verified, and GC content or exon-length distributions are not matched. The goal is not to generate transcripts that would necessarily be spliced in vivo, but to create misleading annotations that are superficially consistent with simple splice-site rules and could plausibly appear in a reference annotation. This tests whether reconstruction tools primarily follow the supplied annotation or instead use read evidence to recover the true splice pattern.

Changes in Manuscript: We have revised the term to "structurally plausible" rather than "biologically plausible" and clarified in the Methods (Section 4.4, lines 644–648) that artificial splice sites preserve canonical motifs but do not guarantee functional splicing.

3.9 Comment 9: Parameter selection justification

> **Reviewer Comment**
>
> It's not clear how certain parameter values were selected. Moreover, the sensitivity of results to those choices is not clear. This could be better argued/explained. For example:
>
> - **3.a.** How were the window sizes for TSS, TTS, and CAGE decided, and what is the sensitivity of the results to those parameters?
> - **3.b.** Genes in the TUSCO gene set are medium to highly expressed across tissues (Section 2.1); is there value in lowering the minimum expression threshold to capture lower-yet-consistently-expressed genes, and how are LRS method evaluation results affected?
> - **3.c.** Splice evidence can require as little as 1 supporting read for a junction in gene selection in the TUSCO benchmarking framework (Section 4.1); is there value in more stringency? Should this part of the pipeline handle known confounders such as read stacks / artifacts and require junction read support from multiple positions? How would this change the results?
> - **3.d.** How were the different thresholds for mouse vs human RNAseq data decided?

Response

We conducted systematic sensitivity analyses for each major parameter category. Procedure is described in the new Methods section 4.2, and results are consolidated in new Supplementary Fig. 1

3.a. TSS/CAGE window size validation:
The ±300 bp window was chosen to balance sensitivity against inter-gene collision. Using refTSS v4.1 on single-isoform genes, we computed two complementary metrics as a function of window size (Supplementary Fig. 1a–b). At 300 bp, the human data show 96.6% sensitivity (fraction of genes whose annotated TSS falls within the window) against only 2.1% collision (fraction with a neighboring gene's TSS within range); mouse figures are comparable (93.8% vs 4.3%). The 75th-percentile peak count at this threshold is 2, meaning roughly one quarter of genes (those with complex promoter architecture) are excluded. Because the collision rate remains below 5%, these exclusions reflect genuine alternative TSSs rather than technical artifacts from nearby genes. Narrower windows sacrifice sensitivity; wider ones admit too many alternative promoters.

3.b. Expression thresholds:
We swept four configurations spanning strict to lenient expression criteria:

Human	Prevalence	Median RPKM	Min Expr	Splice Purity	n_genes
Strict	0.99	1.0	0.10	0.90	36
Current	0.95	0.5	0.05	0.90	46
Lenient	0.75	0.2	0.02	0.90	64
More Lenient	0.55	0.1	0.01	0.90	79

Mouse	Prevalence	Median RPKM	Min Expr	Splice Purity	n_genes
Strict	0.95	1.0	0.10	0.90	30
Current	0.90	0.5	0.05	0.90	32
Lenient	0.70	0.2	0.02	0.90	40
More Lenient	0.50	0.1	0.01	0.90	58

Here, *Prevalence* is the fraction of tissues where a gene must be expressed; *Median RPKM* and *Min Expr* set expression floors; and *Splice Purity* (held at 0.90) is the fraction of tissues whose splice ratio falls below threshold.
We then benchmarked each configuration against SIRV spike-in controls across six LRGASP library preparation workflows (Supplementary Fig. 1c–d):

Species	Config	n_genes	Median TUSCO evaluation FN%	Median SIRV FN%	Difference
Human	Strict	36	9.7%	3.7%	+6.0%
Human	Current	46	18.5%	3.7%	+14.8%
Human	Lenient	64	28.9%	3.7%	+25.2%
Human	More Lenient	79	36.1%	3.7%	+32.4%
Mouse	Strict	30	20.0%	4.5%	+15.5%
Mouse	Current	32	23.4%	4.5%	+18.9%
Mouse	Lenient	40	31.2%	4.5%	+26.7%
Mouse	More Lenient	58	49.1%	4.5%	+44.6%

False negative rates from TUSCO evaluation exceed SIRV baselines because endogenous genes are subject to RNA degradation and variable expression that synthetic spike-ins bypass. The "Current" configuration adds 10 genes (human) or 2 genes (mouse) relative to "Strict" with only a moderate rise in FN; moving further to "Lenient" nearly doubles FN rates, indicating that those additional genes are simply not reliably detected at typical sequencing depths. As demonstrated elsewhere in the manuscript, FN in the "Current" set correlates with sequencing depth, not threshold choice. Lowering thresholds would include genes that remain undetected regardless.

3.c. Splice junction read support:
The "≥1 read" floor exists only to prevent divide-by-zero when $\mu = 0$; the operative filter is the relative threshold $T = \alpha\mu$, where μ is mean annotated-junction coverage and α is 0.01 (human) or 0.05 (mouse). In recount3, μ is typically large (minimum values among final multi-exon genes in the TUSCO gene set are 5.1×10^5 (human) and 9.8×10^5 (mouse)), so the 1-read floor never determines inclusion.
We verified this by plotting Empirical Cumulative Distribution Function (ECDF) of the ratio $\max(\text{novel})/\mu$ for genes in the TUSCO gene set versus single-isoform genes outside the TUSCO

gene set (Supplementary Fig. 1e). Genes in the TUSCO gene set cluster well below the chosen α (95th percentiles: 0.0075 human, 0.032 mouse), whereas excluded genes show markedly higher ratios. Read stacks and other artifacts are further mitigated by requiring junctions to be fully intragenic, strand-consistent, and $\geq$80 bp.

3.d. Human vs. mouse threshold differences:
Mouse and human expression distributions differ in both the number of available RNA-seq reference tracks and the typical absolute RPKM scale. Therefore, we did not enforce a single universal RPKM cutoff, but instead selected species-specific "current" expression thresholds from the same parameter sweep (Supplementary Fig. 1c) to balance gene count against a sharp rise in false negatives under more lenient settings.
Importantly, these species-specific cutoffs still target *comparably high-expression* loci: using AlphaGenome tissue-median RPKM values across all available AlphaGenome expression tracks, the median gene in the TUSCO gene set falls at the 90.3th percentile in human and the 96.5th percentile in mouse (Supplementary Fig. 1d). This high-expression-tail placement supports reliable detectability in benchmarking datasets, consistent with the observation that all genes in the TUSCO gene set are detected in at least one LRGASP dataset (Fig. 3d).

Changes in Manuscript: We have added a Parameter Sensitivity Analysis section to the Methods (lines 559–573), and point to results in line 118 as: "Parameter sensitivity analysis is provided in Supplementary Fig. 1" :

> **Parameter sensitivity analysis.** To validate the robustness of the TUSCO gene set selection pipeline, we conducted systematic sensitivity analyses across four key parameter categories. For TSS/CAGE window size, we evaluated windows from 100 bp to 500 bp using refTSS peak data, measuring sensitivity (fraction of annotated TSSs captured) against inter-gene collision rate; the selected $\pm$300 bp window achieves 96.6% sensitivity in human with only 2.1% collision. For expression thresholds, we swept four configurations (strict to lenient) using LRGASP evaluation data and benchmarked false-negative rates against SIRV controls, selecting parameters that balance gene count against acceptable FN increase. Species-specific thresholds were calibrated to target comparably high-expression percentiles (90th–97th percentile) based on AlphaGenome-predicted tissue-median RPKM in both human and mouse (Supplementary Fig. 1d). For splice junction filtering, we used AlphaGenome-predicted junction read counts to verify that the relative threshold $T = \alpha\mu$ (where μ is mean annotated-junction coverage and α is 0.01 for human, 0.05 for mouse) effectively discriminates genes in the TUSCO gene set from excluded single-isoform genes.

3.10 Comment 10: Cosine similarity interpretation

> **Reviewer Comment**
>
> In the TUSCO evaluation vs SIRV comparison of Section 2.3, it is not intuitive how good the concordance is from the reported cosine similarity. Is there an available negative control e.g. biased protocol or short-read-only protocol, or even "reasonable" simulated result that would illustrate a baseline cosine similarity for this high-dimensional comparison?

Response

Cosine similarity measures the angular similarity between two 6-dimensional metric vectors (Sn, nrPre, Redundancy^{-1}, 1-FDR, PDR, rPre). A value of 1.0 indicates identical relative performance profiles, while values near 0 indicate orthogonal profiles. Values above 0.99 suggest that TUSCO evaluation and SIRVs report virtually identical relative performance, indicating both benchmarks assess the same underlying pipeline quality. Moderate concordance in the 0.94–0.99 range suggests some divergence, potentially due to TUSCO evaluation's sensitivity to factors SIRVs do not capture, such as RNA degradation.
While we did not include a formal negative control (e.g., deliberately biased protocol), we observed that a randomly shuffled metric vector would yield cosine similarity ∼0.5–0.7. The lowest observed cosine similarity (0.9493 for WTC11 cDNA ONT) still indicates strong concordance.

Changes in Manuscript: The following interpretation guidance has been added to Section 2.3 (Results, lines 206–209):

> Values approaching 1 indicate strong agreement in evaluating transcript reconstruction performance but do not necessarily reflect absolute performance quality; as a practical guide, values >0.95 indicate strong concordance, while values <0.90 would suggest meaningful benchmark divergence.

3.11 Comment 11: TP coverage variability

> **Reviewer Comment**
>
> The TP coverages reported in Section 2.5 are highly variable (via the +), is there more to say about when it is higher vs lower?

Response

The large "±" values reflect dispersion across *individual reads* supporting transcripts labeled TP. We find that TP per-read coverage is systematically **higher and more stable for short reference transcripts**, and **lower and much more variable for long reference transcripts**. This is consistent with length-dependent truncation (RNA degradation, incomplete cDNA synthesis/capture), which becomes more likely as transcript length increases.
Concretely, stratifying Iso-Seq TP reads by the reference transcript exonic length shows:

- **Human (WTC11 PacBio cDNA):** for TP reads mapping to transcripts ≤1 kb, mean per-read coverage is 95.7% with SD 8.3%; for 2–3 kb it is 100.2% ± 1.7%; but for longer (3–5 kb) it drops to 80.4% ± 30.5%.
- **Mouse (ES PacBio cDNA):** TP coverage remains high for most length bins (e.g., 3–5 kb: 97.1% ± 8.1%), but declines for very long transcripts (>5 kb: 88.4% ± 21.0%).

Thus, TP per-read coverage is highest when reads are near full-length (shorter transcripts and/or high-integrity molecules) and lowest when truncation is prevalent (longer transcripts and degraded/partially captured molecules), which inflates the SD even when the median coverage remains close to 1 for many bins.

Changes in Manuscript: We have added the following to Section 2.5 (Results, lines 328–329):

> These patterns are consistent with partial transcript coverage effects (Supplementary Fig. 6).

The length-stratified statistics are provided in Supplementary Fig. 6, showing that TP per-read coverage is highest for shorter transcripts and declines for longer transcripts due to length-dependent truncation.

3.12 Comment 12: Per-read coverage definition

> **Reviewer Comment**
>
> It is not clear why per-read coverage is defined and used as in Section 2.5.

Response

In Section 2.5, we investigate why some TUSCO gene set loci are classified as TP (full match, including boundaries) while others are PTP (no novel junction chain, but shifted TSS/TTS). For Iso-Seq data, many PTPs are expected from 5'/3' truncated molecules resulting from RNA degradation or incomplete capture: reads can support the correct junction chain but fail to span the true transcript ends, yielding boundary-shifted models unless transcript ends are refined.
To quantify this end-completeness, we define per-read coverage as $f = L_{\text{read}}/L_{\text{exonic}}$, where L_{exonic} is the total exonic length of the true reference isoform and L_{read} is the observed Iso-Seq read length (PacBio cDNA; FLNC). With this definition, $f \approx 1$ indicates a near full-length molecule, $f \ll 1$ indicates truncation, and f can exceed 1 when a read is longer than the reference isoform (e.g., extra UTR/poly(A) sequence). Because transcript models are constructed directly from the supporting reads, truncated reads lead to incomplete transcript models with shifted TSS/TTS boundaries—precisely the PTP outcome. Consistent with this, TP reads show high coverage (median 98.3% in human), whereas PTP reads show substantially lower coverage (median 31.7% in human), confirming that PTP arises from truncated input molecules rather than algorithmic errors.

Changes in Manuscript: The per-read coverage metric is defined in Section 2.5 (Results, lines 320–325):

> We quantified per-read coverage $f = \frac{L_{\text{read}}}{L_{\text{exonic}}}$ for each read supporting TP or PTP calls in the TUSCO gene set, where L_{read} is the observed Iso-Seq read length (PacBio cDNA; FLNC) and L_{exonic} is the total exonic length of the true reference isoform. Thus, $f \approx 1$ indicates near full-length support, whereas low f indicates 5'/3' truncation expected under RNA degradation or incomplete capture (and f can exceed 1 when a read is longer than the reference isoform).

3.13 Comment 13: Code availability

> **Reviewer Comment**
>
> *Haven't reviewed the code at this stage.*

Response

We note that the reviewer has not reviewed the code at this stage. The code and data are available as described in the "Data and code availability" section of the manuscript.

4 Reviewer #4

Reviewer's Overall Assessment

In the manuscript by Tianyuan Liu et al. authors introduce the Transcriptome Universal Single-isoform COntrol (TUSCO), which is a curated reference set of genes lacking alternative isoforms. The aim is to provide an internal ground truth for benchmarking purposes. The TUSCO benchmarking framework evaluates precision by identifying transcripts deviating from reference annotations and assesses sensitivity by verifying detection completeness in human and mouse samples.

The study is very extensive and detailed what makes that the following the narration is very tricky, although the article is well structure and very well written. Simply the complexity of topic is high and it is difficult to provide an easy solution here.

I am confident that the TUSCO benchmarking framework will become a widely used approach for benchmarking. It has its own limitations which were also specified by the authors. The main is that it is "restricted to single-isoform loci and thus cannot evaluate transcript reconstruction accuracy within highly complex multi-isoform genes". To overcome this limitations authors introduce a set of artificial isoforms derived from genes in the TUSCO gene set. It is a nice trick which allows to assess some properties of the pipelines but still is not perfect. Authors are aware of it and are planning further work in this direction.

Response

We thank the reviewer for seeing the value of our work and for encouraging us to continue development of the TUSCO benchmarking framework.

4.1 Comment 1: Section 2.4 - Correlation interpretation

Reviewer Comment

It is stated: "A pronounced negative correlation (Pearson r $= -0.8$, p $= 1.7 \times 10^{-3}$; n $= 12$ pipelines) was observed, indicating that as sequencing coverage and read length increase, the number of false negatives decreases."

An inspection of Supplementary Fig. 4 reveals that in fact we do have a two clearly separated groups of points. One from ONT and second by PB which are strongly separated both by False Negative rate as well as composite metric. The true source of observed phenomenon here is hidden in sequencing technology differences and is not fully proven here. Thus further statements coming from this part are overinterpretation.

Response

We thank the reviewer for pointing this out. We agree that the earlier correlation framing was an overinterpretation: in this LRGASP subset, FN rates are primarily separated by sequencing platform/protocol (ONT versus PacBio).

Changes in Manuscript: We revised the text in Section 2.4 (lines 258–261) accordingly and updated Supplementary Fig. 4 to a grouped raincloud plot summarizing FN rates by platform and library type. The revised text reads:

> "Consistent with our hypothesis that FN reflects protocol limitations, FN rates across these same LRGASP datasets are dominated by platform/protocol choice (PacBio cDNA lowest; ONT higher; Supplementary Fig. 4)."

4.2 Comment 2: Section 2.5 - TUSCO-novel evaluation

> **Reviewer Comment**
>
> The capacity of reconstruction tools to detect unannotated isoforms is assessed with used of introduced TUSCO-novel reference set. The authors note that tools which heavily rely on annotation fail to recover new isoforms as defined in TUSCO-novel. That is true, but what was not highlighted or noted is the fact that preferred by authors Iso-Seq+SQANTI3 is not way better. It has marginal drop of performance between "reference annotation" and TUSCO-novel due to the fact that is dealing with "reference annotation" noticeably worse than other approaches. Thus ability of TUSCO-novel to evaluate novel isoforms discovery is not that remarkable I would say.

Response

The reviewer makes an interesting observation. Summarizing Fig. 4b by F1 score, Bambu achieves near-perfect performance under reference annotation (F1: 99.5%, range 97.9–100.0 in human), while Iso-Seq + SQANTI3 ML scores lower (F1: 54.5%). However, under TUSCO-novel—where the reference annotation is deliberately incorrect—Iso-Seq + SQANTI3 ML's F1 remains stable (57.1%), while Bambu's F1 drops dramatically to 37.3% (range 30.5–53.1) (Supplementary Table 3). This demonstrates that Iso-Seq + SQANTI3 ML's moderate but stable performance is largely independent of annotation quality.

The key insight from TUSCO-novel is not absolute performance but annotation dependence. Bambu and StringTie2 are strongly reference-guided and therefore highly dependent on reference annotation; in settings where gene-level annotation is largely complete but isoform-level annotation is missing or incorrect, this reliance can limit their ability to detect genuinely novel isoforms. Consistent with this, when the annotation at loci in the TUSCO gene set is replaced by an incorrect splice pattern, Bambu and StringTie2 fail substantially to recover the true isoforms despite read support. Iso-Seq + SQANTI3 ML, by contrast, performs consistently regardless of annotation status. The lower baseline for Iso-Seq arises because the pipeline lacks TSS/TTS refinement against the reference, resulting in more PTP calls even when the junction chain is correct. This reduces TP rates but makes the pipeline inherently reference-agnostic.

The practical implication of this observation is that in scenarios where reference annotations may be incomplete or misleading (novel tissues, disease states, non-model organisms), a pipeline with lower but *stable* performance may be more desirable than one that performs well only when guided by complete annotations but degrades sharply otherwise. In such settings, maintaining very high RNA quality becomes especially important to minimize the risk of PTP.

Changes in Manuscript: We have revised Section 2.5 (lines 308–311) to explicitly acknowledge this trade-off:

> "By contrast, Iso-Seq + SQANTI3 ML ($n = 1$ per species) showed stable F1 regardless of annotation (human: 54.5% to 57.1%; mouse: 51.0% to 48.9%) and reported 0 FP under TUSCO-novel (Supplementary Table 3)."

4.3 Comment 3: Section 2.6 - Replicate number recommendations

> **Reviewer Comment**
>
> It is stated: "... an optimal design includes at least three biological replicates (Fig. 5b; Supplementary Table 4). By contrast, analyses of SIRV spike-in controls yielded highly consistent concordance metrics across 1–5 replicates, providing little guidance on replicate number optimization (Supplementary Fig. 6)."
>
> An inspection of Fig. 5 together with Supplementary Fig. 6 reveals that there are no drastic differences across 2-5 replicates for both approaches. Thus statement that for TUSCO a good choice is 3+ while for SIRV it is hard to say is overinterpretation. Also here I do not understand what Supplementary Table 4 is adding here.

Response

We agree that the differences among 2–5 replicates are modest for both TUSCO evaluation and SIRVs, and our original phrasing may have overstated the guidance provided.
The most substantial improvement occurs between 1 and 2 replicates, with FDR dropping from 32.9% to 12.2% in brain and from 34.8% to 19.0% in kidney. Beyond 2 replicates, gains are incremental. The key difference between the two benchmarks is that TUSCO metrics do change with replicate number (albeit modestly after 2 replicates), while SIRVs are essentially invariant. This demonstrates that TUSCO evaluation captures real experimental variation that SIRVs miss, even if the absolute differences are small. Regarding Supplementary Table 4, the per-sample FLNC read counts provide the depth context needed to interpret these replicate effects and to generalize them: they allow readers to compare their expected per-sample yield to ours and, under a fixed sequencing budget, to evaluate the trade-off between sequencing deeper versus adding replicates.

Changes in Manuscript: We have revised Section 2.6 (lines 367–375) to present findings more objectively:

> "Collectively, these results indicate that the TUSCO evaluation effectively reveals the effect of replication on transcript detection accuracy. The largest precision gain occurred between one and two replicates, with additional replicates (3–5) yielding incremental improvements at this sequencing depth (Supplementary Table 4). Thus, for this experimental setup (mouse brain/kidney, PacBio cDNA, FLAIR), at least two biological replicates substantially improve precision, with diminishing returns thereafter; additional replicates can still provide modest sensitivity gains (notably in kidney) via consensus boundary refinement (Fig. 5b). By contrast, analyses of SIRV spike-in controls yielded highly consistent concordance metrics across 1–5 replicates, providing little guidance on replicate number optimization (Supplementary Fig. 7)."

5 Reviewer #5

Reviewer's Overall Assessment

I co-reviewed this manuscript with one of the reviewers who provided the listed reports. This is part of the Nature Communications initiative to facilitate training in peer review and to provide appropriate recognition for Early Career Researchers who co-review manuscripts.

No comments provided.